

Student Name: _____

Student Number: _____

Marker Name: _____

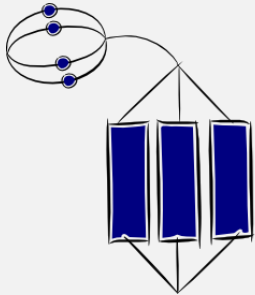
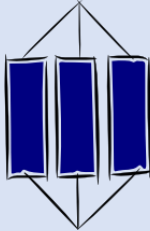
Moderator Name: _____

Total Mark: _____

This is the marking rubric for the PGM project.

For masters student 30 marks = 100%

For honours students 24 marks = 100%

	Extended Abstract (5)	Relational (4)	Multi-structural (3)	Uni-structural (2)	Pre-structural (0-1)
					

(1) Problem Statement and Domain Definition This criterion evaluates how well the student defines the problem statement, establishes its significance within the field, and provides context within the relevant domain. It assesses the originality and impact of the problem statement.	Student demonstrates a profound understanding of the problem's significance in the broader field, creatively links it to existing knowledge, and proposes novel ideas or approaches that have the potential to significantly contribute to the field.	Student effectively establishes the importance of the problem statement within the field, draws connections to relevant literature, and suggests some innovative elements.	Student presents a clear problem statement and mentions its significance within the field, but the connections to existing knowledge are somewhat limited.	Student provides a problem statement but lacks depth in relating it to the broader field or existing knowledge.	Student fails to define a clear problem statement or connect it to the chosen field.
(2) Data Handling and Ethical Considerations: This criterion combines ethical considerations with data preprocessing and feature engineering. It evaluates the ethical awareness of the student, the effectiveness of data preprocessing techniques, and the relevance of feature engineering.	Student shows a deep understanding of ethical considerations, employs advanced data preprocessing techniques creatively, and conducts innovative feature engineering that reflects a high ethical standard.	Student effectively applies ethical principles to data handling, employs relevant data preprocessing techniques, and conducts feature engineering with ethical considerations in mind.	Student demonstrates competency in data preprocessing and feature engineering, considering ethical aspects, but without relevant or creative approaches.	Student performs basic data preprocessing and feature engineering with some consideration for ethics.	Student does not address ethical considerations and struggles with data preprocessing and feature engineering.
(3) PGM Selection and Application:	Student creatively selects and justifies complex PGM models, skillfully	Student effectively selects and justifies PGM models, manages model complexity	Student competently selects and justifies PGM models, manages	Student selects and justifies PGM models, but may	Student does not effectively select,

This criterion covers multiple aspects of applying PGMs, including model selection, justification for chosen models, management of model complexity, experiment design, and the quality of model evaluation.	manages model complexity, designs innovative experiments, and critically evaluates models with advanced techniques.	well, designs experiments appropriately, and conducts thorough model evaluations.	model complexity adequately, designs experiments, and conducts standard model evaluations.	struggle with model complexity, experiment design, or model evaluation.	justify, or apply PGM models.
(4) Understanding of PGM Concepts: This criterion assesses the depth of the student's understanding of key Probabilistic Graphical Models (PGM) concepts, including graphical model types, inference methods, and probabilistic reasoning for the chosen field.	Based only on the submission, student demonstrates a deep and comprehensive understanding of advanced PGM concepts, potentially extending these concepts with innovative ideas.	Based only on the submission, student shows a strong grasp of PGM concepts and their interconnections, applying them effectively to the problem.	Based only on the submission, student understands and applies foundational PGM concepts effectively to the problem but may not explore deeper connections or extensions.	Based only on the submission, student demonstrates a basic understanding of PGM concepts but struggles with their application.	Based only on the submission, student lacks a fundamental understanding of PGM concepts.
(5) Inference and Results Analysis:	Student conducts advanced inference, analyzes results	Student effectively conducts inference, interprets results thoroughly, and performs	Student competently conducts inference, interprets results, and	Student struggles with conducting inference, result	Student does not effectively perform inference, interpret

This criterion assesses the student's ability to perform meaningful inference using PGMs and analyze the results. It includes interpretation of findings and conducting robustness and sensitivity analyses.	comprehensively, and employs innovative techniques for robustness and sensitivity analysis.	robustness and sensitivity analysis effectively.	performs basic robustness and sensitivity analysis.	interpretation, or robustness and sensitivity analysis.	results, or conduct robustness and sensitivity analysis.
(6) Code Quality and Documentation: This criterion evaluates the quality of code implementation, adherence to coding best practices, and the clarity of documentation. It also encourages creativity and innovation in coding approaches.	Student demonstrates exceptional coding skills, produces well-documented, innovative, and efficient code, showing creativity and innovation.	Student produces high-quality, well-documented code that meets assignment guidelines effectively.	Student produces functional, documented code that adheres to guidelines but may lack advanced creativity.	Student produces basic code with minimal documentation or struggles to adhere to assignment guidelines.	Student fails to produce functional code or documentation according to guidelines.